Increasing and decreasing functions

1

a Decreasing **b** Increasing

2

a 1 **b** $(1, \infty)$ **c** $(-\infty, 1)$

3 $(-\infty, 2)$

4

a

i $(-\infty, -2) \cup (2, \infty)$ **ii** $(-2, 2)$

b

i $(-1, 0) \cup (1, \infty)$ **ii** $(-\infty, -1) \cup (0, 1)$

5

a $(0, 0)$ **b** $(0, 0)$ **c** Increasing

6

a $(-1, 0)$ **b** Decrease **c** Increase

Zeros of functions

7

a

i $(4, 0), (-2, 0)$ **ii** $4, -2$

b

i $(2, 0), (-4, 0)$ **ii** $2, -4$

c

i $(3, 0)$ **ii** 3

8

a -3 **b** $3, -1$ **c** $5, 3$

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End behavior

9

a $(-4,0),(0,0)$ **b** $(0,0)$ **c** ∞ **d** ∞

10

a The function does not have x-intercepts. **b** $(0,1)$

c ∞

11

a

i $(6,0),(0,0)$ **ii** $(0,0)$ **iii** Increasing **iv** Increasing

b

i $(-6,0),(6,0)$ **ii** $(0,36)$ **iii** Decreasing **iv** Decreasing

c

i $(-1,0),(1,0)$ **ii** $(0,1)$ **iii** Decreasing **iv** Increasing

d

i $(3,0)$ **ii** $(0,7)$ **iii** Decreasing **iv** Increasing

e

i $(-4,0),(0,0),(4,0)$ **ii** $(0,0)$

iii Increasing **iv** Increasing

f

i $(-2,0),(-1,0),(3,0),(4,0)$ **ii** $(0,4)$

iii Increasing **iv** Increasing

g

i $(-4,0),(3,0),(1,0)$ **ii** $(0,-6)$

iii Decreasing **iv** Increasing

h

i The function does not have x-intercepts.

ii $(0,1)$

iii Increasing

iv Decreasing

i

i $(0,0),(4,0),(0,3)$ **ii** $(0,0)$

iii Decreasing **iv** Decreasing

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j

i $(0,0)$ ii $(0,0)$ iii Increasing iv Decreasing

Asymptotes

12

a 0 b $y=0$ c True

13

a 1 b $y=1$ c True

14

a

i $-\infty$ ii ∞ iii 0 iv 0

b $x=0$

c $y=0$

15

a

i ∞ ii $-\infty$ iii 4 iv 4

b $x=5$

c $y=4$

16

a

i ∞ ii ∞ iii 0 iv 0

b $x=1$

c $y=0$

17

a

i 1 ii 1

b $y=1$ c $(-\infty,\infty)$

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